

School of Computer Science and Artificial Intelligence**Lab Assignment -6**

Name of Student : Chatharaboina sagar
Enrollment No. : 2303A51522
Batch No. : 22

Task -1: Use an AI tool to generate eligibility logic.

Prompt-->

write a program to generate eligibility logic for conditional loops to generate python code to check voting eligibility based on age citizenship

Expected Output:

- AI-generated conditional logic.
- Correct eligibility decisions.
- Explanation of conditions.

```
lab6.py > ...
1  #---Task--1
2  '''write a program to generate eligibility logic for conditional loops
3  to generate python code to check voting eligibility based on age citizenship'''
4  def check_voting_eligibility(age, citizenship):
5      if age >= 18 and citizenship.lower() == 'yes':
6          return "You are eligible to vote."
7      else:
8          return "You are not eligible to vote."
9  #by the input from the user
10 age = int(input("Enter your age: "))
11 citizenship = input("Are you a citizen? (yes/no): ")
12 result = check_voting_eligibility(age, citizenship)
13 print(result)
```

Code-:

output:

```
Coding/lab6.py"
Enter your age: 29
Are you a citizen? (yes/no): yes
You are eligible to vote.
```

Task-2:
Use an AI
tool to

process strings using loops.

Prompt:-

Write a program to string using the loops to generate the python code to count vowels and consonants in a given string

Expected Output:

- AI-generated string processing logic.
- Correct counts.
- Output verification.

Code:

```
lab6.py > count_vowels_consonants
15  #---Task--2
16  '''Write a program to string using the loops
17  to generate the python code to count vowels and consonants in a given string'''
18  def count_vowels_consonants(input_string):
19      vowels = "aeiouAEIOU"
20      vowel_count = 0
21      consonant_count = 0
22
23      for char in input_string:
24          if char.isalpha():
25              if char in vowels:
26                  vowel_count += 1
27              else:
28                  consonant_count += 1
29
30      return vowel_count, consonant_count
31  #by the input from the user
32  input_string = input("Enter a string: ")
33  vowels, consonants = count_vowels_consonants(input_string)
34  print(f"Vowels: {vowels}, Consonants: {consonants}")
35
```

Output:

```
Coding/lab6.py"
Enter a string: sagar
Vowels: 2, Consonants: 3
```

Justification :

- **Loop** is used to check all characters in the string.
- **is alpha()** is used to count only alphabets (ignore spaces/numbers).
- Program gives correct number of vowels and consonants.

Explanation :

This program takes a string input and checks each character using a loop.

If the character is a vowel it increases vowel count, otherwise it increases consonant count.

Task-->3 Use an AI tool to generate a complete program using classes,

loops, and conditionals.

Prompt:

Generate python code to a complete program using classes and loops and conditional to manage for a simple library management system using classes and loops, conditional statements

Expected Output:

- Complete AI-generated program.
- Review of AI suggestions quality.
- Short reflection on AI-assisted coding experience.

Code:

```
lab6.py > Library
36 #---Task--3
37 """Generate python code to a complete program using classes and loops and conditonal
38 to manage for a simple library management system using classes and loops, conditional satatements
39 """
40
41 class Book:
42     def __init__(self, title, author):
43         self.title = title
44         self.author = author
45         self.is_available = True
46
47     def __str__(self):
48         return f"{self.title} by {self.author} - {'Available' if self.is_available else 'Checked Out'}"
49
50 class Library:
51     def __init__(self):
52         self.books = []
53
54     def add_book(self, book):
55         self.books.append(book)
56         print(f'Book "{book.title}" added to the library.')
57
58     def display_books(self):
59         if not self.books:
60             print("No books in the library.")
61             return
62         for idx, book in enumerate(self.books, start=1):
63             print(f"{idx}. {book}")
64
65     def check_out_book(self, book_index):
66         if 0 <= book_index < len(self.books):
67             book = self.books[book_index]
68             if book.is_available:
69                 book.is_available = False
70                 print(f'You have checked out "{book.title}"')
71             else:
72                 print(f'Sorry, "{book.title}" is already checked out.')
73         else:
74             print("Invalid book index.")
75
76     def return_book(self, book_index):
77         if 0 <= book_index < len(self.books):
78             book = self.books[book_index]
79             if not book.is_available:
80                 book.is_available = True
```

```

lab6.py > Library
49 class Library:
75     def return_book(self, book_index):
76         if not book.is_available:
77             if not book.is_available:
78                 book.is_available = True
79                 print(f'You have returned "{book.title}"')
80             else:
81                 print(f'"{book.title}" was not checked out.')
82         else:
83             print("Invalid book index.")
84
85 def main():
86     library = Library()
87     while True:
88         print("\nLibrary Management System")
89         print("1. Add Book")
90         print("2. Display Books")
91         print("3. Check Out Book")
92         print("4. Return Book")
93         print("5. Exit")
94         choice = input("Enter your choice (1-5): ")
95
96         if choice == '1':
97             title = input("Enter book title: ")
98             author = input("Enter book author: ")
99             book = Book(title, author)
100             library.add_book(book)
101         elif choice == '2':
102             library.display_books()
103         elif choice == '3':
104             library.display_books()
105             book_index = int(input("Enter the book index to check out: ")) - 1
106             library.check_out_book(book_index)
107         elif choice == '4':
108             library.display_books()
109             book_index = int(input("Enter the book index to return: ")) - 1
110             library.return_book(book_index)
111         elif choice == '5':
112             print("Exiting the program.")
113             break
114         else:
115             print("Invalid choice. Please try again.")
116
117 if __name__ == "__main__":
118     main()

```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter your choice (1-5): 2
No books in the library.

Library Management System
1. Add Book
2. Display Books
3. Check Out Book
4. Return Book
5. Exit
Enter your choice (1-5): 1
Enter book title: ONE DAY YOU WILL BE WELL SETTLED
Enter book author: SAGAR
Book "ONE DAY YOU WILL BE WELL SETTLED" added to the library.

Library Management System
1. Add Book
2. Display Books
3. Check Out Book
4. Return Book
5. Exit
Enter your choice (1-5): 4
1. ONE DAY YOU WILL BE WELL SETTLED by SAGAR - Available
Enter the book index to return: 3
Invalid book index.

Library Management System
1. Add Book
2. Display Books
3. Check Out Book
4. Return Book
Library Management System
1. Add Book
2. Display Books
3. Check Out Book
4. Return Book
2. Display Books
3. Check Out Book
4. Return Book
3. Check Out Book
4. Return Book
4. Return Book
5. Exit
Enter your choice (1-5): 5
Exiting the program.
```

class.

Prompt:

Generate python code to generate an attendance management class.

Generate a Python class to mark and display student attendance using loops and conditional statements

Expected Output:

- AI-generated attendance logic.
- Correct display of attendance.
- Test cases.

Code:

```

119  #---Task---4
120  '''Generate python code to generate an attendance management class.
121  Generate a Python class to mark and display student
122  attendance using loops and conditional statements.'''
123  class Attendance:
124      def __init__(self):
125          self.attendance_record = {}
126
127      def mark_attendance(self, student_name, present):
128          self.attendance_record[student_name] = present
129          status = "present" if present else "absent"
130          print(f"Marked {student_name} as {status}.")
131
132      def display_attendance(self):
133          if not self.attendance_record:
134              print("No attendance records available.")
135              return
136          print("Attendance Record:")
137          for student, present in self.attendance_record.items():
138              status = "Present" if present else "Absent"
139              print(f"{student}: {status}")
140
141  def main():
142      attendance = Attendance()
143      while True:
144          print("\nAttendance Management System")
145          print("1. Mark Attendance")
146          print("2. Display Attendance")
147          print("3. Exit")
148          choice = input("Enter your choice (1-3): ")
149
150          if choice == '1':
151              student_name = input("Enter student name: ")
152              present_input = input("Is the student present? (yes/no): ").strip().lower()
153              present = True if present_input == 'yes' else False
154              attendance.mark_attendance(student_name, present)
155          elif choice == '2':
156              attendance.display_attendance()
157          elif choice == '3':
158              print("Exiting the program.")
159              break
160          else:
161              print("Invalid choice. Please try again.")
162
163  if __name__ == "__main__":
164      main()

```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Attendance Management System
1. Mark Attendance
2. Display Attendance
3. Exit
Enter your choice (1-3): 2
Attendance Record:
sagar: Present

Attendance Management System
1. Mark Attendance
2. Display Attendance
3. Exit
Enter your choice (1-3): 2
Attendance Record:
sagar: Present

Attendance Management System
1. Mark Attendance
2. Display Attendance
3. Exit
Enter your choice (1-3): 2
Attendance Record:
sagar: Present
Attendance Record:
sagar: Present
sagar: Present

Attendance Management System
1. Mark Attendance
2. Display Attendance
3. Exit
3. Exit
Enter your choice (1-3): 3
Enter your choice (1-3): 3
Exiting the program.
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Coding> █
```

Task-5:

Use an AI tool to complete a navigation menu.

Prompt:

Generate python code to complete a navigation menu.

Generate a Python program using loops and conditionals to simulate an ATM menu navigation system

Expected Output:

- AI-generated menu logic.
- Correct option handling.
- Output verification.

Code:

```
lab6.py > ...
163
164 #---Task--5
165 '''Generate python code to complete a navigation menu.
166 Generate a Python program using loops and conditionals
167 to simulate an ATM menu navigation system.'''
168 def atm_menu():
169     balance = 1000 # Initial balance
170     while True:
171         print("\nATM Menu")
172         print("1. Check Balance")
173         print("2. Deposit Money")
174         print("3. Withdraw Money")
175         print("4. Exit")
176         choice = input("Enter your choice (1-4): ")
177
178         if choice == '1':
179             print(f"Your current balance is: ${balance}")
180         elif choice == '2':
181             amount = float(input("Enter amount to deposit: "))
182             if amount > 0:
183                 balance += amount
184                 print(f"${amount} deposited successfully.")
185             else:
186                 print("Invalid amount. Please enter a positive value.")
187         elif choice == '3':
188             amount = float(input("Enter amount to withdraw: "))
189             if 0 < amount <= balance:
190                 balance -= amount
191                 print(f"${amount} withdrawn successfully.")
192             else:
193                 print("Invalid amount or insufficient balance.")
194         elif choice == '4':
195             print("Exiting the ATM menu. Thank you!")
196             break
197         else:
198             print("Invalid choice. Please try again.")
199 if __name__ == "__main__":
200     atm_menu()
201
```


Output:

```
ATM Menu
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice (1-4): 2
Enter amount to deposit: 5000
$5000.0 deposited successfully.

ATM Menu
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice (1-4): 3
Enter amount to withdraw: 2000
$2000.0 withdrawn successfully.

ATM Menu
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice (1-4): 1
Your current balance is: $4000.0

ATM Menu
Enter your choice (1-4): 1
Your current balance is: $4000.0

ATM Menu
Your current balance is: $4000.0

ATM Menu

ATM Menu
ATM Menu
1. Check Balance
2. Deposit Money
2. Deposit Money
3. Withdraw Money
4. Exit
4. Exit
Enter your choice (1-4): 4
Exiting the ATM menu. Thank you!
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Coding> █
```